

LUMING TANG

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EDUCATION

Cornell University, Department of Computer Science Jan 2019 - Present
Ph.D. in Computer Science
Advisor: Prof. Bharath Hariharan
Research Interests: Computer Vision and Machine Learning

Tsinghua University, Department of Physics Aug 2014 - Jul 2018
B.S. in Mathematics and Physics
Second Major in Economics

PUBLICATIONS AND MANUSCRIPTS

(* indicates equal contribution)

Luming Tang, Nataniel Ruiz, Qinghao Chu, Yuanzhen Li, Aleksander Holynski, David E. Jacobs, Bharath Hariharan, Yael Pritch, Neal Wadhwa, Kfir Aberman, Michael Rubinstein. “RealFill: Reference-Driven Generation for Authentic Image Completion”, *Tech Report* 2023.

Luming Tang, Menglin Jia, Qianqian Wang, Cheng Perng Phoo, Bharath Hariharan. “Emergent Correspondence from Image Diffusion”, in *Conference on Neural Information Processing Systems* (NeurIPS 2023).

Chen-Hsuan Lin*, Jun Gao*, **Luming Tang***, Towaki Takikawa*, Xiaohui Zeng*, Xun Huang, Karsten Kreis, Sanja Fidler, Ming-Yu Liu, Tsung-Yi Lin. “Magic3D: High-Resolution Text-to-3D Content Creation”, in *Conference on Computer Vision and Pattern Recognition* (CVPR 2023), **Highlight**.

Menglin Jia*, **Luming Tang***, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, Ser-Nam Lim. “Visual Prompt Tuning”, in *European Conference on Computer Vision* (ECCV 2022).

Davis Wertheimer*, **Luming Tang***, Bharath Hariharan. “Few-Shot Classification with Feature Map Reconstruction Networks”, in *Conference on Computer Vision and Pattern Recognition* (CVPR 2021).

Luming Tang, Davis Wertheimer, Bharath Hariharan. “Revisiting Pose-Normalization for Fine-Grained Few-Shot Recognition”, in *Conference on Computer Vision and Pattern Recognition* (CVPR 2020).

Luming Tang, Yexiang Xue, Di Chen, Carla P. Gomes. “Multi-Entity Dependence Learning with Rich Context via Conditional Variational Auto-encoder”, in *AAAI Conference on Artificial Intelligence* (AAAI 2018).

Zhongdao Wang*, **Luming Tang***, Xihui Liu, Zhuliang Yao, Shuai Yi, Jing Shao, Junjie Yan, Shengjin Wang, Hongsheng Li, Xiaogang Wang. “Orientation Invariant Feature Embedding and Spatial Temporal Regularization for Vehicle Re-identification”, in *International Conference on Computer Vision* (ICCV 2017).

Luming Tang, Boyang Deng, Haiyu Zhao, Shuai Yi. “Hierarchical Deep Recurrent Architecture for Video Understanding”, in *CVPR 2017 Workshop on Youtube-8M Large-Scale Video Understanding*.

EXPERIENCE

Google Research, Research Intern May 2023 - Present
Hosts: Neal Wadhwa, Nataniel Ruiz, Miki Rubinstein

- Worked on RealFill: reference-driven generation for authentic image completion.

NVIDIA Research, Research Intern

May 2022 - Feb 2023

Mentor: Tsung-Yi Lin

- Worked on Magic3D: high-resolution text-to-3D generation and editing with image diffusion prior.

Waymo, Perception Intern

May 2021 - Nov 2021

Mentors: Shiwei Sheng, Andy Tsai, Ruichi Yu

- Worked on end-to-end data-driven 3D multi-object tracking with Transformer.

Microsoft Research Asia, Research Intern

Sep 2018 - Dec 2018

Mentor: David Wipf

- Worked on training high-resolution VAE and enabling high-quality image generation.

Cornell University, Research Intern

Jun 2017 - Sep 2017

Advisor: Carla Gomes

- Worked on structured multi-entity distribution modeling with a VAE-based algorithm.

SenseTime, Research Intern

Dec 2016 - Jun 2017

Mentor: Shuai Yi

- Worked on vehicle Re-ID with orientation-invariant embedding and spatio-temporal regularization.
- Worked on large-scale video understanding with a hierarchical deep recurrent architecture.

Tsinghua University, Research Assistant

Sep 2016 - Jun 2018

Advisor: Zhiyuan Liu

- Worked on relation extraction with attention mechanism.

TEACHING ASSISTANTS

CS 4787 Principles of Large-Scale Machine Learning

Spring 2019

CS 2110 OOP and Data Structures

Summer 2019

CS 6670 Graduate Computer Vision

Fall 2019

SELECTED AWARDS

Magic3D named among Top 10 NVIDIA Research Highlights of 2023

Jan 2024

NeurIPS'23 Scholar Award

Oct 2023

CVPR'21 Outstanding Reviewer

Jun 2021

Star of Tomorrow (awarded for distinguished internship), Microsoft Research Asia

Dec 2018

Distinguished Academic Innovation Award, Department of Physics, Tsinghua (2/100)

Oct 2017

Academic Talent Program Scholarship, Tsinghua

Dec 2014

First Prize in Chinese Physics Olympiad Competition, ranked 11-th in Henan Province

Sep 2013

ACADEMIC SERVICES

Reviewer: CVPR'21-24, ICCV/ECCV'21-23, ICML'23-24, NeurIPS'23, ICLR'24, TPAMI-SI, IJCV

PC Member: AAAI'23

SKILLS

Python, PyTorch, TensorFlow, JAX, $\text{\LaTeX}$